AIDCF

AI Data Centre Facilities Assurance Framework

AI DATA CENTRE ASSURANCE REFERENCE ARCHITECTURE

Version 1.0

A capability-centred, system-of-systems architecture connecting AI service outcomes to facilities, engineered infrastructure, compute platforms, operations, evidence, failure paths, critical gates and accountable decisions

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Model the capability. Expose the dependency. Trace the failure. Verify the safeguard. Preserve uncertainty. Bound the decision.

Document control

AIDCF-013 bridges the AIDCF standards stack and deployable assurance implementations.

Element	Architecture requirement
Purpose	Define layers, viewpoints, entities, systems, interfaces, patterns, evidence points, failure paths and graph mappings.
Primary subject	Defined AI Capability Instance and required Service Outcomes.
Scope	Site, physical, power, thermal, environment, compute, storage, network, operations, resilience, supply chain, security, evidence and governance.
Uses	Assessment scoping, AITG graphs, digital twins, schemas, controls, evidence, dashboards and profiles.
Boundary	Not a vendor design, engineering code, operating procedure, certification or testing authority.
Safety	No switching, intrusion or disruption is authorised.

Executive thesis

An AI data centre is represented as a capability-supporting system of systems. A conformant architecture traces Service Outcomes through functions, systems, assets, interfaces, resources, dependencies, states, failure paths and safeguards to evidence, controls, critical gates and bounded decisions.

Canonical chain

Capability → Outcome → Function → Service Chain → System → Asset/Interface/Resource → Dependency/Fault Domain → State → Failure → Safeguard → Evidence → Control → Finding/UNKNOWN → Gate → Score → Decision → Report/Reassessment

Part I - Authority and scope

AIDCF-013 organises existing AIDCF objects into deployable views and patterns.

Element	Architecture requirement
Semantic authority	AIDCF-003 owns canonical entities and relationships.
Failure authority	AIDCF-004 owns threat, hazard and propagation semantics.
Integration authority	AIDCF-005 governs AIDCF-AITG handoffs.
Assurance authority	AIDCF-007 through AIDCF-012 govern controls, evidence, assessment, decisions, programs and reporting.
Vendor neutrality	Functions and requirements precede products.
No universal blueprint	Tailor by capability, site, technology, state, consequence and decision.
Change control	Extensions cannot silently weaken the baseline.

Part II - Architecture principles

Principles constrain all implementations.

Element	Architecture requirement
Capability first	Start from service outcomes, not inventory.
Function before product	Keep requirements vendor-neutral.
System of systems	Connect facility, digital, operational and external services.
Dependency explicitness	Record direct, indirect, shared, external and recovery reliance.
State awareness	Model normal, maintenance, degraded, emergency, recovery and transition.
Failure informed	Represent credible paths and propagation.
Fault-domain clarity	Substantiate independence and common modes.
Evidence attachment	Attach evidence or UNKNOWN to material assertions.
Gate readiness	Design evidence paths for applicable gates.
Human approval	Automation may propose; accountable review approves.

Part III - Conformance levels

Progressive conformance retains common semantics.

Element	Architecture requirement
AR0	Vocabulary aligned.
AR1	Capability and outcomes mapped.
AR2	Infrastructure mapped.
AR3	Dependency and fault domains complete.
AR4	Controls, evidence, scenarios and gates attached.
AR5	Graph conformant and versioned.
AR6	Continuing-assurance sources and triggers defined.

Part IV - Architecture layers

Ten connected layers form the model.

Element	Architecture requirement
L0	Decision and governance.
L1	Capability and outcomes.
L2	Functional service chain.
L3	Logical platform.
L4	Physical IT.
L5	Facility engineering.
L6	External ecosystem.
L7	Operations and resilience.
L8	Assurance objects.
L9	Decision, reporting and reassessment.

Part V - Required viewpoints

Each view is a projection of one controlled model.

Element	Architecture requirement
Capability	Required when material; omission requires rationale.
Functional	Required when material; omission requires rationale.
System	Required when material; omission requires rationale.
Physical	Required when material; omission requires rationale.
Dependency	Required when material; omission requires rationale.
Fault-domain	Required when material; omission requires rationale.
State	Required when material; omission requires rationale.
Capacity	Required when material; omission requires rationale.
Failure	Required when material; omission requires rationale.
Safeguard	Required when material; omission requires rationale.
Control/evidence	Required when material; omission requires rationale.
Decision	Required when material; omission requires rationale.
Graph	Required when material; omission requires rationale.
Reporting	Required when material; omission requires rationale.

Part VI - Capability architecture

The model is anchored to a precise capability record.

Element	Architecture requirement
stable identity/owner/lifecycle/version	Required
mission and users	Required
workload model	Required
demand model	Required
Service Outcomes	Required
tolerances	Required
criticality	Required
architecture baseline	Required
scope/boundary	Required
reassessment triggers	Required

Service Outcome specification

Every outcome is an assessable object.

Element	Architecture requirement
outcomeId	Required
measurable statement	Required
conditions/state	Required
tolerance/duration	Required
critical functions	Required
dependencies	Required
safeguards	Required
evidence floor	Required
critical gates	Required
owner	Required
reassessment triggers	Required

Reference Service Chain - Training

ingest → storage → fabric → accelerators → checkpoint → output

Element	Architecture requirement
Flow	ingest → storage → fabric → accelerators → checkpoint → output
Objects	Functions, systems, interfaces, resources, states, owners and providers.
Dependencies	Internal, shared, external and recovery.
Failure analysis	Modes, paths, safeguards, residual state and recovery.
Assurance	Controls, evidence, findings, UNKNOWN, gates and decisions.
Boundary	Excluded services remain explicit.
Validation	Reconcile logical, physical and observed states.

Reference Service Chain - Inference

ingress → route → preprocess → model service → compute → response

Element	Architecture requirement
Flow	ingress → route → preprocess → model service → compute → response
Objects	Functions, systems, interfaces, resources, states, owners and providers.
Dependencies	Internal, shared, external and recovery.
Failure analysis	Modes, paths, safeguards, residual state and recovery.
Assurance	Controls, evidence, findings, UNKNOWN, gates and decisions.
Boundary	Excluded services remain explicit.
Validation	Reconcile logical, physical and observed states.

Reference Service Chain - Fine-tuning

dataset → transfer → training → checkpoint → evaluation → registry

Element	Architecture requirement
Flow	dataset → transfer → training → checkpoint → evaluation → registry
Objects	Functions, systems, interfaces, resources, states, owners and providers.
Dependencies	Internal, shared, external and recovery.
Failure analysis	Modes, paths, safeguards, residual state and recovery.
Assurance	Controls, evidence, findings, UNKNOWN, gates and decisions.
Boundary	Excluded services remain explicit.
Validation	Reconcile logical, physical and observed states.

Reference Service Chain - Sovereign

authorised ingress → controlled processing → protected storage → approved egress

Element	Architecture requirement
Flow	authorised ingress → controlled processing → protected storage → approved egress
Objects	Functions, systems, interfaces, resources, states, owners and providers.
Dependencies	Internal, shared, external and recovery.
Failure analysis	Modes, paths, safeguards, residual state and recovery.
Assurance	Controls, evidence, findings, UNKNOWN, gates and decisions.
Boundary	Excluded services remain explicit.
Validation	Reconcile logical, physical and observed states.

Reference Service Chain - Shared platform

tenant intake → scheduling → shared infrastructure → monitoring

Element	Architecture requirement
Flow	tenant intake → scheduling → shared infrastructure → monitoring
Objects	Functions, systems, interfaces, resources, states, owners and providers.
Dependencies	Internal, shared, external and recovery.
Failure analysis	Modes, paths, safeguards, residual state and recovery.
Assurance	Controls, evidence, findings, UNKNOWN, gates and decisions.
Boundary	Excluded services remain explicit.
Validation	Reconcile logical, physical and observed states.

Reference Service Chain - Recovery

detection → decision → containment → alternate capacity → restore → validate

Element	Architecture requirement
Flow	detection → decision → containment → alternate capacity → restore → validate
Objects	Functions, systems, interfaces, resources, states, owners and providers.
Dependencies	Internal, shared, external and recovery.
Failure analysis	Modes, paths, safeguards, residual state and recovery.
Assurance	Controls, evidence, findings, UNKNOWN, gates and decisions.
Boundary	Excluded services remain explicit.
Validation	Reconcile logical, physical and observed states.

GOV Domain Reference Architecture - Governance

Canonical profile for governance.

Element	Architecture requirement
Objects	authority; ownership; policy; review
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	charters; decisions
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-13; CG-14
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

CAP Domain Reference Architecture - Capability

Canonical profile for capability.

Element	Architecture requirement
Objects	capability; workload; outcome; demand; envelope
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	register; SLO; plan
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-01; CG-02
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

DEP Domain Reference Architecture - Dependencies

Canonical profile for dependencies.

Element	Architecture requirement
Objects	dependency; provider; shared service; concentration
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	graph; contract; route
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-03; CG-12
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

PHY Domain Reference Architecture - Physical

Canonical profile for physical.

Element	Architecture requirement
Objects	site; building; room; zone; rack; barrier
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	drawing; inspection; access
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-02; CG-04
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

POW Domain Reference Architecture - Power

Canonical profile for power.

Element	Architecture requirement
Objects	utility; transformer; UPS; generator; distribution
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	SLD; telemetry; tests
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-05
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

THM Domain Reference Architecture - Thermal

Canonical profile for thermal.

Element	Architecture requirement
Objects	chiller; pump; CRAH; CDU; loop; leak detection
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	P&ID; telemetry; tests
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-06
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

ENV Domain Reference Architecture - Environment

Canonical profile for environment.

Element	Architecture requirement
Objects	sensor; meter; water; waste; hazard zone
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	telemetry; permits
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-02; CG-04
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

CMP Domain Reference Architecture - Compute

Canonical profile for compute.

Element	Architecture requirement
Objects	accelerator; host; rack; cluster; scheduler; firmware
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	topology; health; tests
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-08
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

STO Domain Reference Architecture - Storage

Canonical profile for storage.

Element	Architecture requirement
Objects	tier; controller; metadata; replication; checkpoint
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	integrity; restore
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-08; CG-10
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

NET Domain Reference Architecture - Network

Canonical profile for network.

Element	Architecture requirement
Objects	fabric; switch; optic; cable; OOB; time
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	topology; telemetry
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-08
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

OPS Domain Reference Architecture - Operations

Canonical profile for operations.

Element	Architecture requirement
Objects	operator; procedure; alarm; incident; change
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	work orders; tickets
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-07; CG-13; CG-14
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

RES Domain Reference Architecture - Resilience

Canonical profile for resilience.

Element	Architecture requirement
Objects	fault domain; plan; exercise; crisis team
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	scenario; exercise
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-09; CG-10
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

SUP Domain Reference Architecture - Supply Chain

Canonical profile for supply chain.

Element	Architecture requirement
Objects	supplier; OEM; spare; support; update source
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	contract; provenance
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-03; CG-13
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

SEC Domain Reference Architecture - Security

Canonical profile for security.

Element	Architecture requirement
Objects	identity; privilege; zone; gateway; control plane
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	config; logs; tests
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-07; CG-15
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

EVA Domain Reference Architecture - Evidence/Assurance

Canonical profile for evidence/assurance.

Element	Architecture requirement
Objects	evidence; assertion; result; finding; gate; decision
Relationships	IMPLEMENTS; CONNECTS_TO; DEPENDS_ON; LOCATED_IN; CONTROLLED_BY; MONITORED_BY; PROTECTED_BY.
Boundary	Owned, operated, third-party, shared and excluded.
States	Normal, maintenance, degraded, emergency, recovery and transition.
Capacity	Rated, available, reserved, consumed, headroom and constraint.
Fault domains	Redundancy, common modes and containment.
Evidence	manifest; workpaper
Controls	Applicable AIDCF-007 controls.
Critical gates	CG-11; CG-12
Outputs	View, dependency map, scenarios and evidence points.
Reassessment	Configuration, capacity, provider, incident or evidence change.

System Architecture Card - Incoming power

Reusable record for incoming power.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	utility; feeders; protection; transfer
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

System Architecture Card - Generation/fuel

Reusable record for generation/fuel.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	generators; starting; fuel; controls
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

System Architecture Card - UPS/storage

Reusable record for ups/storage.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	inverter; bypass; batteries
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

System Architecture Card - Distribution

Reusable record for distribution.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	switchgear; busway; PDU; protection
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

System Architecture Card - Air cooling

Reusable record for air cooling.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	chiller; tower; pump; CRAH; containment
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

System Architecture Card - Liquid cooling

Reusable record for liquid cooling.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	CDU; loops; manifolds; fluid; leak detection
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

System Architecture Card - Compute cluster

Reusable record for compute cluster.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	nodes; accelerators; scheduler; firmware
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

System Architecture Card - Storage platform

Reusable record for storage platform.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	tiers; controllers; metadata; replication; checkpoint
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

System Architecture Card - Network fabrics

Reusable record for network fabrics.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	scale-up; scale-out; service; storage; OOB; optics
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

System Architecture Card - BMS/EPMS

Reusable record for bms/epms.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	sensor; controller; gateway; HMI; historian
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

System Architecture Card - Operations centre

Reusable record for operations centre.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	people; monitoring; work; incident; change
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

System Architecture Card - Provider ecosystem

Reusable record for provider ecosystem.

Element	Architecture requirement
Purpose	Required function and linked outcomes.
Components/interfaces	utility; telco; water; fuel; OEM; logistics
Capacity/envelope	Demand, available capacity, headroom and constraints.
States	Normal through transition.
Dependencies	Upstream, shared, external and recovery.
Fault domains	Independence and common modes.
Failure	Modes and propagation.
Safeguards	Prevent, detect, contain, correct and recover.
Evidence	Design, configuration, operation and validation.
Controls/gates	Applicable controls and gates.
Owners	Design, operation, maintenance and evidence.
Triggers	Versioned reassessment events.

Interface Contract - Electrical

Minimum contract for electrical interfaces.

Element	Architecture requirement
interfaceId/version	Required
endpoints/purpose	Required
properties/units	Required
states/transitions	Required
capacity/performance	Required
authority/change rights	Required
monitoring/source health	Required
failure/safe state	Required
evidence	Required
controls/gates	Required
reassessment	Required

Interface Contract - Thermal

Minimum contract for thermal interfaces.

Element	Architecture requirement
interfaceId/version	Required
endpoints/purpose	Required
properties/units	Required
states/transitions	Required
capacity/performance	Required
authority/change rights	Required
monitoring/source health	Required
failure/safe state	Required
evidence	Required
controls/gates	Required
reassessment	Required

Interface Contract - Fluid

Minimum contract for fluid interfaces.

Element	Architecture requirement
interfaceId/version	Required
endpoints/purpose	Required
properties/units	Required
states/transitions	Required
capacity/performance	Required
authority/change rights	Required
monitoring/source health	Required
failure/safe state	Required
evidence	Required
controls/gates	Required
reassessment	Required

Interface Contract - Network

Minimum contract for network interfaces.

Element	Architecture requirement
interfaceId/version	Required
endpoints/purpose	Required
properties/units	Required
states/transitions	Required
capacity/performance	Required
authority/change rights	Required
monitoring/source health	Required
failure/safe state	Required
evidence	Required
controls/gates	Required
reassessment	Required

Interface Contract - Management

Minimum contract for management interfaces.

Element	Architecture requirement
interfaceId/version	Required
endpoints/purpose	Required
properties/units	Required
states/transitions	Required
capacity/performance	Required
authority/change rights	Required
monitoring/source health	Required
failure/safe state	Required
evidence	Required
controls/gates	Required
reassessment	Required

Interface Contract - Data

Minimum contract for data interfaces.

Element	Architecture requirement
interfaceId/version	Required
endpoints/purpose	Required
properties/units	Required
states/transitions	Required
capacity/performance	Required
authority/change rights	Required
monitoring/source health	Required
failure/safe state	Required
evidence	Required
controls/gates	Required
reassessment	Required

Interface Contract - Physical

Minimum contract for physical interfaces.

Element	Architecture requirement
interfaceId/version	Required
endpoints/purpose	Required
properties/units	Required
states/transitions	Required
capacity/performance	Required
authority/change rights	Required
monitoring/source health	Required
failure/safe state	Required
evidence	Required
controls/gates	Required
reassessment	Required

Interface Contract - Operational handoff

Minimum contract for operational handoff interfaces.

Element	Architecture requirement
interfaceId/version	Required
endpoints/purpose	Required
properties/units	Required
states/transitions	Required
capacity/performance	Required
authority/change rights	Required
monitoring/source health	Required
failure/safe state	Required
evidence	Required
controls/gates	Required
reassessment	Required

Interface Contract - Provider

Minimum contract for provider interfaces.

Element	Architecture requirement
interfaceId/version	Required
endpoints/purpose	Required
properties/units	Required
states/transitions	Required
capacity/performance	Required
authority/change rights	Required
monitoring/source health	Required
failure/safe state	Required
evidence	Required
controls/gates	Required
reassessment	Required

Interface Contract - Assurance

Minimum contract for assurance interfaces.

Element	Architecture requirement
interfaceId/version	Required
endpoints/purpose	Required
properties/units	Required
states/transitions	Required
capacity/performance	Required
authority/change rights	Required
monitoring/source health	Required
failure/safe state	Required
evidence	Required
controls/gates	Required
reassessment	Required

Reference Pattern - Single-site training facility

Implementation pattern for single-site training facility.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Reference Pattern - Dual-site capability

Implementation pattern for dual-site capability.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Reference Pattern - Colocation deployment

Implementation pattern for colocation deployment.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Reference Pattern - Hyperscale campus

Implementation pattern for hyperscale campus.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Reference Pattern - Modular facility

Implementation pattern for modular facility.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Reference Pattern - Liquid-cooled high-density cluster

Implementation pattern for liquid-cooled high-density cluster.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Reference Pattern - Hybrid cloud/on-prem

Implementation pattern for hybrid cloud/on-prem.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Reference Pattern - Sovereign AI environment

Implementation pattern for sovereign ai environment.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Reference Pattern - Shared multi-tenant platform

Implementation pattern for shared multi-tenant platform.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Reference Pattern - Edge inference estate

Implementation pattern for edge inference estate.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Reference Pattern - Brownfield retrofit

Implementation pattern for brownfield retrofit.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Reference Pattern - Recovery-capacity pool

Implementation pattern for recovery-capacity pool.

Element	Architecture requirement
Question	Can the pattern support declared outcomes in defined states?
Boundary	Included and external systems/providers.
Views	Capability, system, dependency, fault domain, state, capacity, failure and assurance.
Evidence	Architecture, configuration, operation, changes and validation.
Gates	Derived from material outcomes and dependencies.
UNKNOWN	Unverified shared/external reliance remains visible.
AITG	Reviewed native objects and handoffs.
Reassessment	Pattern-specific change, incident and expiry triggers.

Operating-state model

Canonical architecture model.

Element	Architecture requirement
NORMAL	Required and evidence-linked.
MAINTENANCE	Required and evidence-linked.
DEGRADED	Required and evidence-linked.
EMERGENCY	Required and evidence-linked.
RECOVERY	Required and evidence-linked.
TRANSITION	Required and evidence-linked.
UNKNOWN	Required and evidence-linked.

Capacity and envelope

Canonical architecture model.

Element	Architecture requirement
Demand	Required and evidence-linked.
Rated	Required and evidence-linked.
Available	Required and evidence-linked.
Reserved	Required and evidence-linked.
Consumed	Required and evidence-linked.
Headroom	Required and evidence-linked.
Constraint	Required and evidence-linked.
Burst	Required and evidence-linked.
Growth	Required and evidence-linked.
UNKNOWN	Required and evidence-linked.

Fault-domain model

Canonical architecture model.

Element	Architecture requirement
FaultDomain	Required and evidence-linked.
RedundancyGroup	Required and evidence-linked.
IndependenceClaim	Required and evidence-linked.
CommonMode	Required and evidence-linked.
ContainmentBoundary	Required and evidence-linked.
BlastRadius	Required and evidence-linked.
RecoveryDomain	Required and evidence-linked.
Evidence	Required and evidence-linked.

Failure-path model

Canonical architecture model.

Element	Architecture requirement
initiating source	Required and evidence-linked.
pre-state	Required and evidence-linked.
failure mode	Required and evidence-linked.
local effect	Required and evidence-linked.
dependency edge	Required and evidence-linked.
next entity	Required and evidence-linked.
safeguard	Required and evidence-linked.
residual state	Required and evidence-linked.
outcome effect	Required and evidence-linked.
evidence/confidence	Required and evidence-linked.
gate effect	Required and evidence-linked.

Safeguard model

Canonical architecture model.

Element	Architecture requirement
identity	Required and evidence-linked.
function	Required and evidence-linked.
subject	Required and evidence-linked.
activation	Required and evidence-linked.
authority	Required and evidence-linked.
dependencies	Required and evidence-linked.
state coverage	Required and evidence-linked.
failure modes	Required and evidence-linked.
evidence	Required and evidence-linked.
control mapping	Required and evidence-linked.

Observability model

Canonical architecture model.

Element	Architecture requirement
metric/event	Required and evidence-linked.
source	Required and evidence-linked.
subject	Required and evidence-linked.
time/clock	Required and evidence-linked.
state context	Required and evidence-linked.
quality/gaps	Required and evidence-linked.
threshold/owner	Required and evidence-linked.
alarm route	Required and evidence-linked.
evidence use	Required and evidence-linked.
blind spot	Required and evidence-linked.

Evidence attachment

Canonical architecture model.

Element	Architecture requirement
assertion	Required and evidence-linked.
evidence classes	Required and evidence-linked.
source authority	Required and evidence-linked.
collection	Required and evidence-linked.
quality floor	Required and evidence-linked.
sufficiency	Required and evidence-linked.
conflict	Required and evidence-linked.
UNKNOWN rule	Required and evidence-linked.
retention	Required and evidence-linked.
reassessment	Required and evidence-linked.

Control attachment

Canonical architecture model.

Element	Architecture requirement
controlId	Required and evidence-linked.
subject	Required and evidence-linked.
applicability	Required and evidence-linked.
dimensions	Required and evidence-linked.
methods	Required and evidence-linked.
evidence	Required and evidence-linked.
result/confidence	Required and evidence-linked.
finding/UNKNOWN	Required and evidence-linked.
gates	Required and evidence-linked.
reassessment	Required and evidence-linked.

Critical-gate architecture

Gate readiness is designed into the architecture.

Gate	Architecture focus
CG-01	Capability/outcome definition
CG-02	Boundary/architecture integrity
CG-03	Critical dependency visibility
CG-04	Life safety/hazardous energy
CG-05	Electrical continuity
CG-06	Thermal continuity
CG-07	Control-system integrity
CG-08	Compute/storage/network integrity
CG-09	Failure containment
CG-10	Recovery readiness
CG-11	Evidence sufficiency
CG-12	Critical UNKNOWN
CG-13	Responsibility/authority
CG-14	Change control
CG-15	Integrated digital authority

Canonical graph nodes

The graph projects the controlled architecture.

Element	Architecture requirement
CapabilityInstance	Stable ID, version, source, status and classification.
ServiceOutcome	Stable ID, version, source, status and classification.
RequiredFunction	Stable ID, version, source, status and classification.
InfrastructureSystem	Stable ID, version, source, status and classification.
AssetInstance	Stable ID, version, source, status and classification.
Interface	Stable ID, version, source, status and classification.
Resource	Stable ID, version, source, status and classification.
Dependency	Stable ID, version, source, status and classification.
FaultDomain	Stable ID, version, source, status and classification.
OperatingState	Stable ID, version, source, status and classification.
FailureScenario	Stable ID, version, source, status and classification.
Safeguard	Stable ID, version, source, status and classification.
EvidenceObject	Stable ID, version, source, status and classification.
ControlAssessment	Stable ID, version, source, status and classification.
GateResult	Stable ID, version, source, status and classification.
AssuranceDecision	Stable ID, version, source, status and classification.

Canonical graph relationships

Edges preserve native meaning and evidence.

Element	Architecture requirement
SUPPORTS_OUTCOME	Directed, scoped, versioned, reviewed and evidence-linked.
REQUIRES_FUNCTION	Directed, scoped, versioned, reviewed and evidence-linked.
IMPLEMENTED_BY	Directed, scoped, versioned, reviewed and evidence-linked.
COMPOSED_OF	Directed, scoped, versioned, reviewed and evidence-linked.
CONNECTS_TO	Directed, scoped, versioned, reviewed and evidence-linked.
DEPENDS_ON	Directed, scoped, versioned, reviewed and evidence-linked.
SHARES_FAULT_DOMAIN_WITH	Directed, scoped, versioned, reviewed and evidence-linked.
LOCATED_IN	Directed, scoped, versioned, reviewed and evidence-linked.
OPERATED_BY	Directed, scoped, versioned, reviewed and evidence-linked.
CONTROLLED_BY	Directed, scoped, versioned, reviewed and evidence-linked.
MONITORED_BY	Directed, scoped, versioned, reviewed and evidence-linked.
PROTECTED_BY	Directed, scoped, versioned, reviewed and evidence-linked.
EXPOSED_TO	Directed, scoped, versioned, reviewed and evidence-linked.
PROPAGATES_TO	Directed, scoped, versioned, reviewed and evidence-linked.
RECOVERS_THROUGH	Directed, scoped, versioned, reviewed and evidence-linked.
EVIDENCED_BY	Directed, scoped, versioned, reviewed and evidence-linked.
ASSESSED_BY	Directed, scoped, versioned, reviewed and evidence-linked.
CONSTRAINS_GATE	Directed, scoped, versioned, reviewed and evidence-linked.
LIMITS_DECISION	Directed, scoped, versioned, reviewed and evidence-linked.
REPORTED_IN	Directed, scoped, versioned, reviewed and evidence-linked.
SUPERSEDES	Directed, scoped, versioned, reviewed and evidence-linked.
INTEGRATION_HANDOFF	Directed, scoped, versioned, reviewed and evidence-linked.

Graph validation rules

Graph implementations fail safely.

ID	Rule
ARA-GV-01	Stable IDs and lineage.
ARA-GV-02	Capability links outcomes.
ARA-GV-03	Outcomes link functions and systems.
ARA-GV-04	Dependencies have direction, conditions and evidence.
ARA-GV-05	Redundancy links fault domains and independence evidence.
ARA-GV-06	Scenarios include states, paths, safeguards and outcomes.
ARA-GV-07	Controls link subjects and evidence.
ARA-GV-08	Gates link mandatory evidence.
ARA-GV-09	Decisions link scope, gates, UNKNOWN, validity and authority.
ARA-GV-10	Inference cannot become fact automatically.
ARA-GV-11	Connectivity does not imply causality/reachability/authority.
ARA-GV-12	AIDCF/AITG decisions remain separate.
ARA-GV-13	Sensitive paths are restricted.
ARA-GV-14	Supersession preserves history.

Required work products

Select applicable products and justify omissions.

Element	Architecture requirement
Capability Context Diagram	Required when material to scope and decision.
Service Chain Map	Required when material to scope and decision.
System Decomposition	Required when material to scope and decision.
Power Assurance View	Required when material to scope and decision.
Thermal/Fluid Assurance View	Required when material to scope and decision.
Compute/Storage/Network View	Required when material to scope and decision.
Dependency Graph	Required when material to scope and decision.
Fault-Domain Map	Required when material to scope and decision.
Operating-State Matrix	Required when material to scope and decision.
Capacity Model	Required when material to scope and decision.
Failure Map	Required when material to scope and decision.
Safeguard Map	Required when material to scope and decision.
Control/Evidence Matrix	Required when material to scope and decision.
Critical-Gate Map	Required when material to scope and decision.
Decision Lineage View	Required when material to scope and decision.
AITG Handoff Package	Required when material to scope and decision.

Template - Capability Record

Human and machine-readable capability record.

Element	Architecture requirement
capabilityId	
owner	
lifecycle	
workload	
outcomes	
demand	
criticality	
baseline	
scope	
version	

Template - Outcome Record

Human and machine-readable outcome record.

Element	Architecture requirement
outcomeId	
statement	
conditions	
tolerance	
functions	
dependencies	
safeguards	
evidence	
gates	
owner	

Template - System Record

Human and machine-readable system record.

Element	Architecture requirement
systemId	
purpose	
type	
components	
interfaces	
capacity	
states	
dependencies	
faultDomains	
owners	
evidence	

Template - Asset Record

Human and machine-readable asset record.

Element	Architecture requirement
assetId	
class	
location	
configuration	
state	
system	
owner	
lifecycle	
evidence	

Template - Interface Record

Human and machine-readable interface record.

Element	Architecture requirement
interfaceId	
endpoints	
type	
purpose	
capacity	
authority	
monitoring	
failure	
evidence	

Template - Dependency Record

Human and machine-readable dependency record.

Element	Architecture requirement
dependencyId	
source	
target	
type	
conditions	
criticality	
substitutability	
evidence	
confidence	

Template - FaultDomain Record

Human and machine-readable faultdomain record.

Element	Architecture requirement
faultDomainId	
members	
source	
boundary	
blastRadius	
safeguards	
evidence	

Template - Scenario Record

Human and machine-readable scenario record.

Element	Architecture requirement
scenarioId	
preState	
source	
steps	
safeguards	
residual	
outcomeEffect	
evidence	
confidence	

Template - Safeguard Record

Human and machine-readable safeguard record.

Element	Architecture requirement
safeguardId	
function	
subject	
activation	
authority	
dependencies	
states	
failureModes	
evidence	

Template - Assertion Record

Human and machine-readable assertion record.

Element	Architecture requirement
assertionId	
subject	
predicate	
value	
conditions	
evidence	
UNKNOWN	
reviewer	
status	

Template - Snapshot Record

Human and machine-readable snapshot record.

Element	Architecture requirement
snapshotId	
capability	
scope	
sourceVersions	
nodes	
edges	
validation	
approval	
supersedes	

Architecture Workflow - New capability

Need → outcomes → boundary → chains → systems → dependencies → states/capacity → scenarios → assurance → review.

Element	Architecture requirement
Flow	Need → outcomes → boundary → chains → systems → dependencies → states/capacity → scenarios → assurance → review.
Entry	Authorised question and owner.
Outputs	Versioned architecture, evidence and validation.
Stop	Unsafe action, unresolved authority, source integrity or competence.
UNKNOWN	Preserve unresolved facts.
Review	Architecture, technical, evidence and independent review.
Trigger	Reassessment and supersession.

Architecture Workflow - Existing facility

Baseline → reconcile → discover → UNKNOWN → map → validate → gates → snapshot.

Element	Architecture requirement
Flow	Baseline → reconcile → discover → UNKNOWN → map → validate → gates → snapshot.
Entry	Authorised question and owner.
Outputs	Versioned architecture, evidence and validation.
Stop	Unsafe action, unresolved authority, source integrity or competence.
UNKNOWN	Preserve unresolved facts.
Review	Architecture, technical, evidence and independent review.
Trigger	Reassessment and supersession.

Architecture Workflow - Material change

Detect → impact → suspend stale claims → update → reassess → supersede.

Element	Architecture requirement
Flow	Detect → impact → suspend stale claims → update → reassess → supersede.
Entry	Authorised question and owner.
Outputs	Versioned architecture, evidence and validation.
Stop	Unsafe action, unresolved authority, source integrity or competence.
UNKNOWN	Preserve unresolved facts.
Review	Architecture, technical, evidence and independent review.
Trigger	Reassessment and supersession.

Architecture Workflow - Incident

Reconstruct path → affected claims/gates → residual state → recovery evidence → reassess.

Element	Architecture requirement
Flow	Reconstruct path → affected claims/gates → residual state → recovery evidence → reassess.
Entry	Authorised question and owner.
Outputs	Versioned architecture, evidence and validation.
Stop	Unsafe action, unresolved authority, source integrity or competence.
UNKNOWN	Preserve unresolved facts.
Review	Architecture, technical, evidence and independent review.
Trigger	Reassessment and supersession.

Architecture Workflow - Continuing assurance

Register sources → detect drift → human triage → reassess → update.

Element	Architecture requirement
Flow	Register sources → detect drift → human triage → reassess → update.
Entry	Authorised question and owner.
Outputs	Versioned architecture, evidence and validation.
Stop	Unsafe action, unresolved authority, source integrity or competence.
UNKNOWN	Preserve unresolved facts.
Review	Architecture, technical, evidence and independent review.
Trigger	Reassessment and supersession.

Architecture Workflow - AITG integration

Validate IDs → shared identity → reviewed handoff → lineage/security → federation.

Element	Architecture requirement
Flow	Validate IDs → shared identity → reviewed handoff → lineage/security → federation.
Entry	Authorised question and owner.
Outputs	Versioned architecture, evidence and validation.
Stop	Unsafe action, unresolved authority, source integrity or competence.
UNKNOWN	Preserve unresolved facts.
Review	Architecture, technical, evidence and independent review.
Trigger	Reassessment and supersession.

Synthetic Example - Training cluster

Illustrative only, not a real design or conclusion.

Element	Architecture requirement
Context	Shared cooling control across redundant trains.
Treatment	Model common mode and CG-06/07/09/10.
Views	Capability, systems, dependencies, fault domains, states/capacity, failure and assurance.
Decision	No positive conclusion without evidence and gate support.
Graph	Versioned nodes/edges with review state.

Synthetic Example - Colocation

Illustrative only, not a real design or conclusion.

Element	Architecture requirement
Context	Customer IT depends on provider facility/operations.
Treatment	Separate boundaries and restricted evidence.
Views	Capability, systems, dependencies, fault domains, states/capacity, failure and assurance.
Decision	No positive conclusion without evidence and gate support.
Graph	Versioned nodes/edges with review state.

Synthetic Example - Dual-site recovery

Illustrative only, not a real design or conclusion.

Element	Architecture requirement
Context	Sites share telecom and identity service.
Treatment	Model shared recovery dependencies and exercises.
Views	Capability, systems, dependencies, fault domains, states/capacity, failure and assurance.
Decision	No positive conclusion without evidence and gate support.
Graph	Versioned nodes/edges with review state.

Synthetic Example - Brownfield retrofit

Illustrative only, not a real design or conclusion.

Element	Architecture requirement
Context	Dense load added to legacy power/air cooling.
Treatment	Reconcile baseline, capacity, thermal envelope and change.
Views	Capability, systems, dependencies, fault domains, states/capacity, failure and assurance.
Decision	No positive conclusion without evidence and gate support.
Graph	Versioned nodes/edges with review state.

Conformance validation

Conformant architectures pass semantic, structural, evidence, safety and lineage tests.

ID	Validation
ARA-VAL-001	Capability/outcomes explicit.
ARA-VAL-002	Functions precede products.
ARA-VAL-003	Scope/external dependencies visible.
ARA-VAL-004	Stable IDs.
ARA-VAL-005	States/transitions represented.
ARA-VAL-006	Capacity claims conditioned/evidenced.
ARA-VAL-007	Fault domains/common modes mapped.
ARA-VAL-008	Dependencies evidenced or UNKNOWN.
ARA-VAL-009	Failure paths link safeguards/outcomes.
ARA-VAL-010	Controls attach subjects.
ARA-VAL-011	Evidence attaches assertions.
ARA-VAL-012	Gates separate from scores.
ARA-VAL-013	Decision lineage complete.
ARA-VAL-014	AIDCF/AITG semantics native.
ARA-VAL-015	Sensitive paths controlled.
ARA-VAL-016	Automation cannot approve facts.
ARA-VAL-017	Snapshots immutable/superseded.
ARA-VAL-018	Safety/engineering boundaries preserved.

Review and release gates

Draft pending multidisciplinary review and pilots.

Reviewer/pilot	Status
Methodology author	Pending
Data-centre architect	Pending
Electrical engineer	Pending
Mechanical/liquid-cooling engineer	Pending
AI/HPC architect	Pending
Storage architect	Pending
Network architect	Pending
Operations/reliability	Pending
OT/security	Pending
Evidence specialist	Pending
Ontology/graph engineer	Pending
Safety/legal/privacy	Pending
Colocation pilot	Pending
High-density cooling pilot	Pending
Hybrid/sovereign pilot	Pending
AITG pilot	Pending
Public release	Pending

Final architecture doctrine

AIDCF-013 makes the assurance subject and dependencies explicit. It is not a vendor design, equipment checklist or substitute for licensed engineering. Where architecture or evidence is incomplete, AIDCF preserves UNKNOWN and limits assurance.